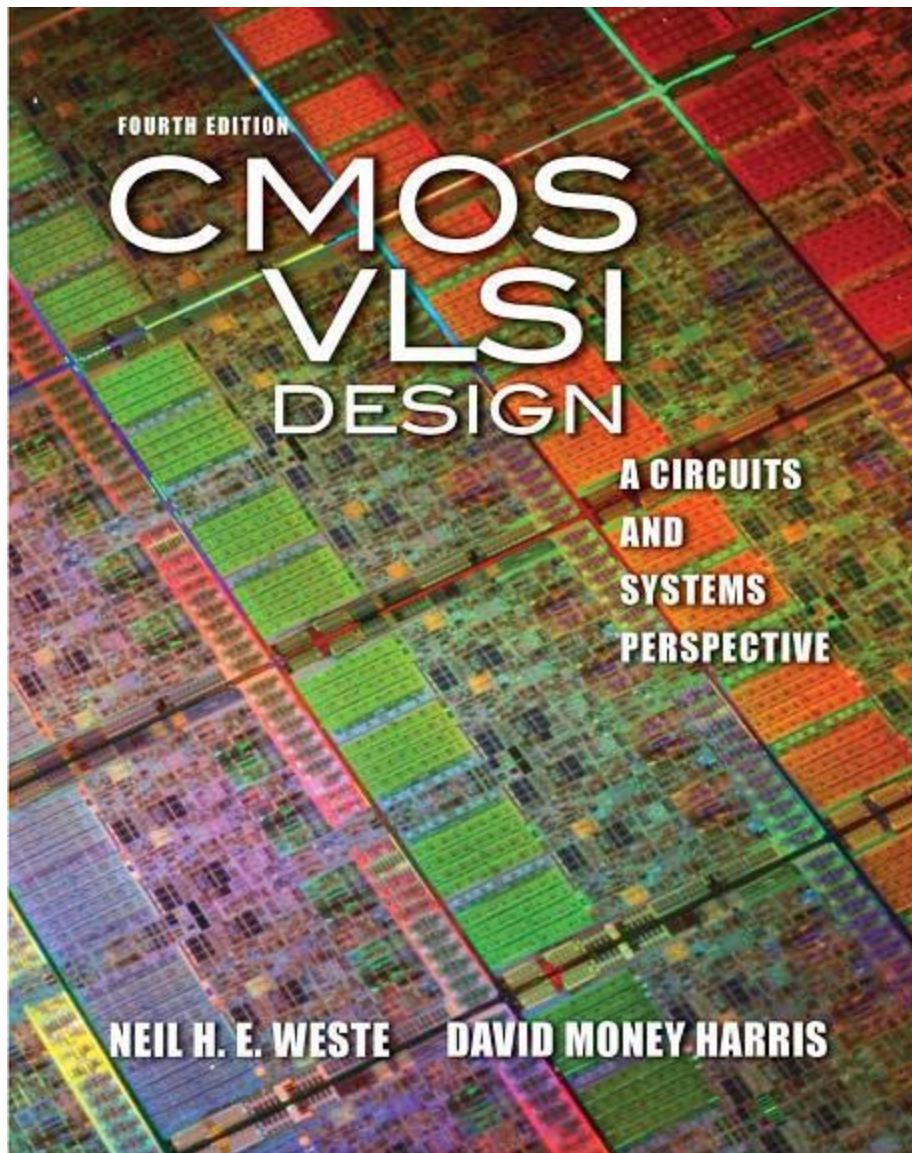




# Chapter 5: Power

# Lecture

- ☐ Power and Energy
- ☐ Static Power
- ☐ Dynamic Power

# Power as a Primary Design Constraint

- ❑ Static CMOS gates are very power-efficient because they dissipate nearly zero power while idle.
- ❑ For much of the history of CMOS design, power was a secondary consideration behind speed and area for many chips.
- ❑ As transistor counts and clock frequencies have increased, power consumption has skyrocketed and now is a primary design constraint.

# Power and Energy

❑ Power is drawn from a voltage source attached to the  $V_{DD}$  pin(s) of a chip.

❑ Instantaneous Power:  $P(t) = I(t)V(t)$

❑ Energy: 
$$E = \int_0^T P(t)dt$$

❑ Average Power: 
$$P_{\text{avg}} = \frac{E}{T} = \frac{1}{T} \int_0^T P(t)dt$$

# Power Dissipation Sources

- ❑  $P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}}$
- ❑ Static power:  $P_{\text{static}} = (I_{\text{sub}} + I_{\text{gate}} + I_{\text{junct}} + I_{\text{contention}})V_{\text{DD}}$ 
  - Subthreshold leakage
  - Gate leakage
  - Junction leakage
  - Contention current
- ❑ Dynamic power:  $P_{\text{dynamic}} = P_{\text{switching}} + P_{\text{shortcircuit}}$ 
  - Switching(charging& discharging) load capacitances
  - Short-circuit current

# Static Power

- ❑ Static power is consumed even when chip is quiescent.
  - Leakage draws power from nominally OFF devices
  - Ratioed circuits burn power in fight between ON transistors

# Charging a Capacitor

- When the gate output rises
  - Energy stored in capacitor is
  - But energy drawn from the supply is

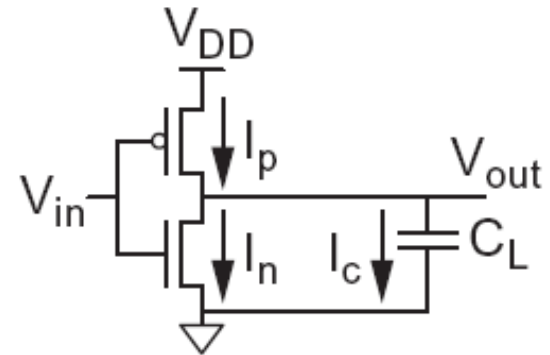
$$E_C = \frac{1}{2} C_L V_{DD}^2$$

$$E_{VDD} = \int_0^\infty I(t) V_{DD} dt = \int_0^\infty C_L \frac{dV}{dt} V_{DD} dt$$

$$= C_L V_{DD} \int_0^{V_{DD}} dV = C_L V_{DD}^2$$

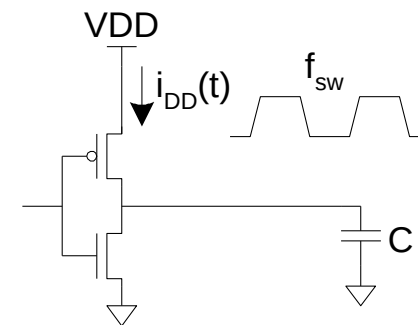
- Half the energy from  $V_{DD}$  is dissipated in the pMOS transistor as heat, other half stored in capacitor

- When the gate output falls
  - Energy in capacitor is dumped to GND
  - Dissipated as heat in the nMOS transistor



# Switching Power

- ❑ Primary dynamic dissipation component is charging the load capacitance.
- ❑ Suppose a load  $C$  is switched between GND and  $V_{DD}$  at an average frequency of  $f_{sw}$
- ❑ Over any given interval of time  $T$ , the load will be charged and discharged  $Tf_{sw}$  times.
- ❑ Current flows from  $V_{DD}$  to the load to charge it.
- ❑ Current then flows from the load to GND during discharge.
- ❑ In one complete charge-discharge cycle, a total charge of  $Q = CV_{DD}$  is thus transferred from  $V_{DD}$  to GND.
- ❑ The switching power dissipation:



$$\begin{aligned} P_{\text{switching}} &= \frac{1}{T} \int_0^T i_{DD}(t) V_{DD} dt \\ &= \frac{V_{DD}}{T} \int_0^T i_{DD}(t) dt \\ &= \frac{V_{DD}}{T} [Tf_{sw} CV_{DD}] \\ &= CV_{DD}^2 f_{sw} \end{aligned}$$



# Activity Factor

- ❑ Suppose the system clock frequency =  $f$
- ❑ Let  $f_{sw} = \alpha f$ , where  $\alpha$  = activity factor
  - If the signal is a clock,  $\alpha = 1$
  - If the signal switches once per cycle,  $\alpha = 1/2$
- ❑ Dynamic power:

$$P_{\text{switching}} = \alpha C V_{DD}^2 f$$

# Short Circuit Current

- ❑ When transistors switch, both nMOS and pMOS networks may be momentarily ON at once
- ❑ Leads to a blip of “short circuit” current.
- ❑  $< 10\%$  of dynamic power if rise/fall times are comparable for input and output
- ❑ We will generally ignore this component

# Dynamic Power Reduction

- ❑  $P_{\text{switching}} = \alpha C V_{DD}^2 f$
- ❑ Try to minimize:
  - Activity factor
  - Capacitance
  - Supply voltage
  - Frequency

# Dynamic Power Example

- ❑ 1 billion transistor chip
  - 50M logic transistors
    - Average width:  $12 \lambda$
    - Activity factor = 0.1
  - 950M memory transistors
    - Average width:  $4 \lambda$
    - Activity factor = 0.02
  - 1.0 V 65 nm process
  - $C = 1 \text{ fF}/\mu\text{m}$  (gate) +  $0.8 \text{ fF}/\mu\text{m}$  (diffusion)
- ❑ Estimate dynamic power consumption @ 1 GHz.  
Neglect wire capacitance and short-circuit current.

# Solution

$$C_{\text{logic}} = (50 \times 10^6)(12\lambda)(0.025 \mu\text{m} / \lambda)(1.8 \text{ fF} / \mu\text{m}) = 27 \text{ nF}$$

$$C_{\text{mem}} = (950 \times 10^6)(4\lambda)(0.025 \mu\text{m} / \lambda)(1.8 \text{ fF} / \mu\text{m}) = 171 \text{ nF}$$

$$P_{\text{dynamic}} = [0.1C_{\text{logic}} + 0.02C_{\text{mem}}](1.0)^2 (1.0 \text{ GHz}) = 6.1 \text{ W}$$

# Static Power Example

- ❑ Revisit power estimation for 1 billion transistor chip
- ❑ Estimate static power consumption
  - Subthreshold leakage
    - Normal  $V_t$ : 100 nA/ $\mu\text{m}$
    - High  $V_t$ : 10 nA/ $\mu\text{m}$
    - High  $V_t$  used in all memories and in 95% of logic gates
  - Gate leakage 5 nA/ $\mu\text{m}$
  - Junction leakage negligible

# Solution

$$W_{\text{normal-}V_t} = (50 \times 10^6)(12\lambda)(0.025 \mu\text{m} / \lambda)(0.05) = 0.75 \times 10^6 \mu\text{m}$$

$$W_{\text{high-}V_t} = \left[ (50 \times 10^6)(12\lambda)(0.95) + (950 \times 10^6)(4\lambda) \right] (0.025 \mu\text{m} / \lambda) = 109.25 \times 10^6 \mu\text{m}$$

$$I_{\text{sub}} = \left[ W_{\text{normal-}V_t} \times 100 \text{ nA}/\mu\text{m} + W_{\text{high-}V_t} \times 10 \text{ nA}/\mu\text{m} \right] / 2 = 584 \text{ mA}$$

$$I_{\text{gate}} = \left[ (W_{\text{normal-}V_t} + W_{\text{high-}V_t}) \times 5 \text{ nA}/\mu\text{m} \right] / 2 = 275 \text{ mA}$$

$$P_{\text{static}} = (584 \text{ mA} + 275 \text{ mA})(1.0 \text{ V}) = 859 \text{ mW}$$